Marcus Keil

Research Software Engineer, University College London

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Professional Summary

Scientific workflow performance analysis and HPC benchmarking define my work, turning system measurements into actionable insights for scientists. For benchmarking, I standardise scientific workflow execution by fixing data and inputs across runs, to create micro-benchmarks that gauge the performance of hardware when running the workflows being developed for the Square-kilometre Array, one of the world's largest radio telescope projects. During my PhD I developed a Bayesian inference tool with a browser-based user interface that used an SQL-backed model caching system to reduce redundant model calculations in order to speed up the inference. The intersection of HPC benchmarking and profiling coupled with software development focused on statistical tool development positions me to understand the needs of developers, as well as the limitations of available hardware.

Work Experience

Research Software Engineer SKA UKSRC — University College London

Computer Science Department — *Aug. 2023 - Present*

- Modify existing benchmarking framework to accommodate SKA compute APIs, to automatically send benchmarking results to SQL database and update Confluence page with benchmarking results.
- Benchmark hardware available to UKSRC both with micro-benchmarks developed from radio astronomer workflows, and synthetic benchmarks. Benchmarks including CPU vs GPU and AMD vs NVIDIA cost-performance comparison. Initial findings of C++ developed FFT benchmark found AMD to have cost-performance of 25% compared to NVIDIA.
- Profile radio astronomer workflows to provide actionable insights for optimisation, and to quantify the improvements they make.

PhD Projects — University College London

Department of Physics and Astronomy — *Dec. 2019 - Aug. 2023*

- Developed UCLCHEMCMC end-to-end, from a Python wrapper around the Fortran chemical modelling code UCLCHEM, to a browser-based UI, with an SQL-backed caching system to avoid redundant modelling while performing a Markov chain Monte Carlo inference with a Bayesian likelihood function.
- Set-up Elasticsearch database with a Kibana dashboard for the healthcare research group as a cross-disciplinary project. Originally intended as a prototype which was subsequently adopted as the primary implementation.
- Development of virtual reality experience in Unity using C#. This project was used at astrophysics and scientific outreach events to promote astrochemistry.

Masters' Project — University of Bonn

Argelander Institute for Astronomy — *Sep. 2018 - Sep. 2019*

- Training a Random Forest Emulator of Dark Matter only simulations to study feature importance for forming galaxy clusters. This work focused on using the importance values to understand how to improve previous analytical prediction methods.

Bachelors' Project University of Oklahoma
Department of Physics and Astronomy *Dec. 2014 - Dec. 2015*

- Grid simulation of planetary nebulae with statistical analysis of results to understand physical parameters of observed objects. Supplemented with an external summer internship at Lehigh University, applying similar statistical methods to astronomical big data for exoplanet candidate identification.

Education

University College London, PhD London, UK
Dec. 2019 - Aug. 2023

- **Subject of Study:** PhD in Astrophysics, Astrochemistry.
- **Thesis Title:** Statistical analyses of Astrochemical Big Data.

University of Bonn, Masters Bonn, Germany
Oct. 2016 - Sep. 2019

- **Subject of Study:** Masters in Astrophysics.
- **Thesis Title:** Application of Machine learning on Large scale structure formation.

University of Oklahoma, Bachelors Norman, OK, USA
Aug. 2013 - Dec. 2015

- **Subject of Study:** Major in Astrophysics, Minor in Mathematics.

Skills

Languages: English (Native), German (Native)
Coding Languages: Python, Fortran, C#, C++, Bash
Frameworks: Tensorflow, Pytorch, Docker, Singularity/Apptainer
Developer Tools: Git, GitHub, Likwid, Prometheus, Linaro Forge, SLURM, PBS, Kibana
Libraries: Scikit-learn, reframe-hpc, cuFFT, rocFFT
Database technologies: SQL, Elasticsearch